

Practical no. 13

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Class :- CO3K – B

Subject :- DSU

Program :-

```
#include<stdio.h>

#include<conio.h>

#include<stdlib.h>

void insert_at_end();

void insert_after();

void delete_node();

void display();

struct Node{

int data;

struct Node*next;

};

struct Node*head=NULL;

void main(){

int choice;

clrscr();

while(1){

printf("\n----MENU----");

printf("\n1.Insert at End\n2.Insert After\n3.Delete Node\n4.Display\n5.Exit");

printf("\nEnter your choice :- ");

scanf("%d",&choice);

switch(choice){

case 1: insert_at_end();

break;
```

```

case 2: insert_after();

        break;

case 3: delete_node();

        break;

case 4: display();

        break;

case 5: exit(0);

        break;

default: printf("\nInvalid Choice!");

        break;

}

}

getch();

}

void insert_at_end(){

struct Node*newNode, *temp;

newNode=(struct Node*)malloc(sizeof(struct Node));

printf("\nEnter data to be inserted at the end :- ");

scanf("%d",&newNode->data);

newNode->next=NULL;

if(head==NULL){

head=newNode;

}

else{

temp=head;

while(temp->next!=NULL){

temp=temp->next;

```

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}

temp->next=newNode;

}

printf("\nNode with data %d is inserted at the end.",newNode->data);

}

void insert_after(){

struct Node*newNode, *temp=head;

int value;

newNode=(struct Node*)malloc(sizeof(struct Node));

printf("\nEnter the value after which to insert :- ");

scanf("%d",&value);

while(temp!=NULL && temp->data!=value){

temp=temp->next;

}

if(temp==NULL){

printf("\nNode with value %d is not found in the list.",value);

}

printf("\nEnter value to be inserted after %d :- ",value);

scanf("%d",&newNode->data);

newNode->next=temp->next;

temp->next=newNode;

printf("\nNode with data %d is successfully inserted after %d",newNode->data,value);

}

void delete_node(){

struct Node*temp=head, *prev=NULL;

int value;

if(head==NULL){

```

```

printf("\nLinked list is empty");
}

printf("\nEnter value to be deleted :- ");

scanf("%d",&value);

if(head->data==value){

head=head->next;

free(temp);

printf("\nNode with data %d is deleted successfully.",value);

}

while(temp!=NULL && temp->data!=value){

prev=temp;

temp=temp->next;

}

prev->next=temp->next;

free(temp);

printf("\nNode with %d is deleted successfully.",value);

}

void display(){

struct Node*temp=head;

if(head==NULL){

printf("\nLinked list is empty");

}

else{

printf("\nElements of linked list are :- ");

while(temp->next!=NULL){

printf("\n%d",temp->data);

temp=temp->next;

```

```
}  
  
}  
  
}
```

OUTPUT :-

```
NeuTroN DOS-C++ 0.77, Cpu speed: max 100% cycles, Frameskip  
-----MENU-----  
1.Insert at End  
2.Insert After  
3.Delete Node  
4.Display  
5.Exit  
Enter your choice :- 1  
  
Enter data to be inserted at the end :- 10  
  
Node with data 10 is inserted at the end.  
-----MENU-----
```

```
-----MENU-----  
1.Insert at End  
2.Insert After  
3.Delete Node  
4.Display  
5.Exit  
Enter your choice :- 2  
  
Enter the value after which to insert :- 10  
  
Enter value to be inserted after 10 :- 20  
  
Node with data 20 is successfully inserted after 10  
-----MENU-----
```

```
-----MENU-----  
1.Insert at End  
2.Insert After  
3.Delete Node  
4.Display  
5.Exit  
Enter your choice :- 3  
  
Enter value to be deleted :- 20  
  
Node with 20 is deleted successfully.  
-----MENU-----
```

```
-----MENU-----  
1.Insert at End  
2.Insert After  
3.Delete Node  
4.Display  
5.Exit  
Enter your choice :- 4  
  
Elements of linked list are :-  
10  
-----MENU-----
```